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United States Patent	12402699
Kind Code	B2
Date of Patent	September 02, 2025
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User-serviceable device

Abstract

The description relates to devices, such as user-serviceable devices. One example can include a chassis defining a cavity accessible through a rear side and a rear cover positioned against the chassis over the rear side. The device can be configured to allow movement between the rear cover and the chassis substantially parallel to the rear side to lock the rear cover against the chassis and to allow movement in an opposite direction to unlock the rear cover and further configured to impart a biasing force on the rear cover substantially perpendicular to the rear side. The device can also include a release that can be engaged through the chassis to cause the movement of the rear cover in the opposite direction to unlock the rear cover and allow the biasing force to move the rear cover substantially perpendicularly to the rear side away from the chassis.

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Appl. No.: 17/740842

Filed: May 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230363504 A1	Nov. 16, 2023

Publication Classification

Int. Cl.: A45C11/00 (20060101)

U.S. Cl.:

CPC **A45C11/00** (20130101); A45C11/002 (20250101); A45C11/003 (20250101);
A45C2200/15 (20130101)

Field of Classification Search

CPC: A45C (11/00); A45C (2011/002); A45C (2011/003); A45C (2200/15); G06F (1/1656);
G06F (1/166); G06F (1/1626)

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Background/Summary

BACKGROUND

(1) The quest for ever thinner devices has led to devices being constructed in a manner that the devices tend to be difficult or impossible to service. This condition diminishes the user experience and can cause user alienation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings illustrate implementations of the concepts conveyed in the present document. Features of the illustrated implementations can be more readily understood by reference to the following description taken in conjunction with the accompanying drawings. Like reference numbers in the various drawings are used wherever feasible to indicate like elements. Further, the left-most numeral of each reference number conveys the FIG. and associated discussion where the reference number is first introduced.
- (2) FIGS. 1A-1C, 2A, 2B, 3B, 3C, 5A, 5B, and 6 are perspective views of example user-serviceable devices in accordance with the present concepts.
- (3) FIG. 3A is an exploded perspective view of an example user-serviceable device in accordance with the present concepts.
- (4) FIGS. 4A-4C are sectional views of example user-serviceable devices in accordance with the present concepts.

SUMMARY

- (5) This patent relates to devices, and more specifically to serviceability of devices. One example can include a chassis defining a front side and a rear side and a volume in between. The volume can be accessible through the rear side. In an operational state, a rear cover can be positioned against the chassis over the rear side. The device can be configured to allow movement between the rear cover and the chassis substantially parallel to the rear side to lock the rear cover against the chassis and to allow movement in an opposite direction to unlock the rear cover for servicing. The device can also be configured to impart a biasing force on the rear cover substantially perpendicular to the rear side. The device can include a release that can be engaged through the chassis to cause the movement of the rear cover to unlock the rear cover and allow the substantially perpendicular biasing force to move the rear cover away from the chassis.

DESCRIPTION

- (6) The present concepts relate to devices, such as computing devices that can include components positioned in a chassis or housing. Many computing devices employ a first portion (e.g., the chassis) and a corresponding second portion (e.g., cover) to form the volume or enclosure that holds the components, such as processors, batteries, etc. For many form factors, such as tablets, notebooks, and/or wearable devices, consumer preferences are toward smaller form factors, especially thinner (e.g., z-dimension constraints) and/or lighter form factors. These demands led to devices that were sealed at the factory, often with the device placed in an alignment die or mold that aligned the first and second portions and pressed them together to provide an aesthetically-pleasing form factor. A special adhesive, such as adhesive tape, was employed in the alignment die to bond the first and second portions. These devices are difficult to service and require special tools, such as heat guns to disassemble the first and second portions. Further, disassembly tends to damage at least the adhesive tape, among other components, such as those to which the adhesive tape is adhered. Thus, reassembly can only be accomplished with special tools and depends on the availability of replacement components, such as adhesive tape.
- (7) Other traditional configurations that offered better serviceability simply included a screw at each corner of the cover that fastened to the chassis. These traditional solutions left exposed screw heads that diminished the appearance of the device. The user could also lack the proper tools for removing the screws and/or lose the screws during the removal and reinstallation process. Covers could be employed over the screwheads to improve the appearance of the device, but those too could be lost.
- (8) In contrast, the present concepts can satisfy user demands relating to device dimensions and appearance while offering a user-serviceable configuration. The user-serviceable configuration can

be disassembled and reassembled without any special tools and without replacing any components (e.g., no alignment dies and/or no sacrificial components). Despite this simplicity, the present concepts can offer simple removal of the cover with any stylus-shaped tool, such as a ball point pen. Reinstallation of the cover can be achieved without any tools simply by aligning the cover over the chassis and pressing downward and then sliding the cover back into the locked configuration.

(9) FIGS. 1A-1C collectively show some of the present concepts relative to an example device **100**, such as a tablet type device. The device can include a chassis **102** that defines a first or front side **104**, a second or rear side **106**, and a volume **107** in between. A display, though not visible, can be positioned on the front side **104**. The rear side can be divided into a first area **108** and a second area **110**. In this example, the first and second areas **108** and **110** can be separated by a kickstand **112** that can be rotatably secured to the chassis **102** by a hinge **113**. The kickstand **112** can be closed (e.g., folded) against the rear side **106** (e.g., against the first area **108**). The kickstand **112** can also be rotated away from the rear side to various angles to support the device in various postures.

(10) The second area **110** can be covered by a rear cover **114**. The rear cover **114** can be removed to access the volume **107** and components **116**, such as batteries, processors, memory/storage, positioned in the volume **107**. (The components are shown in ghost because they would be occluded in these views). The rear cover can be reinstalled to protect the components **116** in the volume **107**.

(11) A release **118** can be positioned on the first area **108**. In this configuration, the release **118** can be hidden when the kickstand **112** is closed and accessible when the kickstand is open. As shown in FIG. 1C, user activation of the release **118**, such as by depressing the release with a stylus **120**, can cause the rear cover **114** to move horizontally along the rear side **106** (e.g., along the xy reference plane) away from the first area **108** as indicated by arrow **122** and then to move up (e.g., away) from the rear side **106** (as indicated by arrow **124**) where it can be lifted away to access the electronic components. Once finished, the user can re-install the rear cover **114** simply by placing it back on the second area of the chassis **102**, pushing down (e.g., perpendicular to the rear side **106**) and sliding it toward the first area **108** (e.g., parallel to the rear side **106**).

(12) FIGS. 2A and 2B are similar to FIGS. 1A and 1B, respectively and can be considered collectively with FIGS. 1A-1C. FIGS. 2A and 2B show latch assembly **202** and ejection assemblies **204**. These assemblies can lie beneath the rear side **106** and would not be visible in these views and thus are shown in ghost. The latch assembly **202** and ejection assemblies **204** can enable easy removal and replacement of the rear cover **114**. These assemblies are discussed in more detail below and are illustrated here to orient the reader.

(13) FIGS. 3A-3C collectively show additional components of device **100**. FIG. 3A is an exploded perspective view of device **100**. FIGS. 3B and 3C are enlarged views of a region of the device. In this case, front side **104** is facing upward, whereas in FIGS. 1A-1C and 2A and 2B, the rear side **106** is facing upward. In this implementation, latch assembly **202** includes cover plate fasteners **302**, a cover plate **304**, pivot fasteners **306**, shuttle springs **308**, pivot rod **310**, latch spring **312**, latch **314**, and/or latch shuttle **316**. These components of the latch assembly **202** can be positioned in a cavity **318** defined by the chassis **102**. FIG. 3B shows these components of the latch assembly **202** assembled on the device but without the cover plate fasteners **302** and the cover plate **304**.

(14) In this configuration, individual ejection assemblies **204** are positioned in the chassis **102** to operate relative to the four corners of the rear cover. An alternative configuration for ejection assembly placement is described below relative to FIG. 6.

(15) In FIG. 3A, as illustrated relative to ejection assembly **204(1)**, the ejection assemblies **204** can include fasteners **320**, a cover plate **322**, ejection springs **324**, a spring rod **326**, and/or a plug **328**. These components of the ejection assembly **204** can be positioned in cavities **330** defined by the chassis **102**. FIG. 3C shows the ejection assembly positioned in the cavity **330**, but without the cover plate **322** and fasteners **320**.

(16) In FIG. 3A, hooks **332** are visible on the underside of the rear cover **114** (e.g., on the side that contacts the rear side **106**). In this configuration, individual hooks **332** are positioned in each corner of the rear cover **114**. Other configurations are contemplated and an alternative configuration is described below relative to FIG. 6. The hooks **332** can be positioned in the cavities **330** and interact with portions of the chassis **102** to releasably retain the rear cover on the chassis. This aspect will be shown and described below relative to FIGS. 4A-4C. The underside of the rear cover can also define a bumper **334**. The bumper **334** can engage the latch assembly **202** as will be shown and described below relative to FIGS. 4A-4C.

(17) FIGS. 4A-4C and 5A and 5B collectively show operational features of device **100**. FIGS. 4A-4C are sectional views as indicated in FIG. 2A that collectively show operation of the latch assembly **202** and the ejection assembly **204** in relation to retaining and releasing the rear cover **114**. In this configuration, chassis **102** defines lips **402** that partially extend over cavities **330** to define a slot **404**. The slot **404** can be considered as a subset of the cavity **330**.

(18) FIG. 4A shows an operational configuration where the rear cover **114** is secured to the chassis **102** by the ejection assembly **204**(1). At this point, the hooks **332** underly and engage the lips **402** of the chassis **102** to retain the rear cover **114** on the chassis **102**. The hooks **332** and the lips **402** provide a technical solution of preventing movement of the rear cover **114** perpendicular to the rear side **106** while allowing seamless movement parallel to the rear side **106** (e.g., the hooks slideably engage the lips).

(19) At this point, the ejection springs **324** have been compressed by the rear cover pushing the plugs **328** downward substantially perpendicular to the rear side **106** (e.g., in the z reference direction). The ejection springs **324** are thus creating a bias on the plugs **328** from the front side **104** toward the rear side **106**. However, the plugs **328** cannot move upward because the hooks **332** are retained under the lips **402**. This engagement between the hooks **332** of the rear cover **114** and the lips **402** of the chassis **102** is maintained by the latch assembly **202**, which is preventing horizontal movement (e.g., movement in the y reference direction substantially parallel to the rear side **106**) of the rear cover **114**. As used herein, 'substantially parallel' can mean+/-about ten degrees from parallel, for example. Similarly, 'substantially perpendicular' can mean+/-about ten degrees from perpendicular, for example.

(20) Looking at the latch assembly **202**, the latch spring **312** is biasing the latch **314** up against the rear side **106** (e.g., the latch **314** is pivoting around the pivot rod **310**). At this point, the latch **314** is biased toward the rear side **106** by the latch spring **312**. This can be seen in both FIGS. 4A and 5A where latch spring **312** is extended and is biasing the latch **314** toward the rear side **106**. The biasing of the latch **314** is causing a latch hook **406** to engage a corresponding latch shuttle hook **408**. As shown in FIG. 5A, the shuttle springs **308** are compressed between the chassis **102** and the latch shuttle **316** and are creating a bias on the latch shuttle away from the latch **314**. However, the latch spring **312** is biasing the latch **314** toward the rear side **106** and the shuttle hook **408** is essentially captured by the latch hook **406** and the rear cover **114** is maintained in the operational position despite the bias imparted by the shuttle springs **308**. Thus, the latch assembly **202** provides a technical solution to selectively convey the bias that is parallel to the rear side **106** and when to block the bias from acting on the rear cover **114**.

(21) Note that shuttle springs **308**, latch spring **312**, and ejection springs **324** are employed in this implementation to create biases between components. Other biasing mechanisms are contemplated. For instance, compressible elastomeric materials can be used that can be compressed in one direction and create an opposing bias in the other opposite direction.

(22) FIG. 4B shows user engagement of release **118** with a stylus **410**. The user engagement can overcome the bias of latch spring **312** and push the release **118** downward toward the front side **104**. This movement causing the latch **314** to pivot around pivot rod **310** and the latch hook **406** to move downwardly a distance in the z reference direction (DZ). This downward movement frees the shuttle hook **408** from the latch hook **406**. At this point, as seen in FIG. 4B, the bias created by the

shuttle springs **316** can now move the latch shuttle **316** away from the latch **314** in the y reference direction by a distance (DY). This y direction movement is conveyed to the rear cover **114** by the latch shuttle **316** acting on the bumper **334**. As such, the rear cover **114** slides parallel to the rear side **106** and the hooks **332** are moved out from under the lips **402**.

(23) As shown in FIG. 4C, once the hooks **332** clear the lips **402**, the bias imparted on the plugs **328** by the ejection springs **324** can move the plugs, and in turn the rear cover **114**, upwardly in the z reference direction (e.g., in a direction from the front side **104** toward and through the rear side **106**). The plugs **328** have now moved upward until they are flush with the rear side **106** and are occupying space previously occupied by the hooks **332**. Further upward movement of the plugs **328** is blocked by contact between the plugs **328** and the chassis **102**. The plugs **328** are blocking the cavities **330** so that foreign materials cannot enter the cavities. The plugs **328** being flush with the rear side **106** can also contribute to an aesthetically pleasing uniform rear side. At this point, the rear cover **114** is no longer constrained and can be lifted off by the user to access components positioned in the chassis. Note that there are no loose components, such as fasteners for the user to keep track of or lose.

(24) Thus, the plugs **328** provide two technical solutions. First the plugs transfer the upward bias force that is perpendicular to the rear surface to the hooks **332** and the rear cover **114**. Second, the plugs **328** can fill the cavities **330** when the rear cover **114** is removed to prevent foreign materials from entering the cavities **330** and potentially interfering with the function of the ejection assemblies **204** when the rear cover **114** is re-installed.

(25) To reinstall the rear cover **114**, the process can be reversed. The user can place the rear cover **114** on the rear side **106** as shown in FIG. 4C. The user then pushes downward to compress the ejection springs **324** until the hooks **332** clear the lips **402** similar to FIG. 4B. The user can then move the rear cover **114** horizontally until the shuttle hooks **408** re-engage the latch hooks **406** similar to FIG. 4A.

(26) Looking at FIGS. 4A-4C again collectively, the latch assembly **202** and the ejection assemblies **204** collectively provide a technical solution that provides a quick and easy access to internal components for the purposes of repair and/or upgrade. The latch assembly **202** can include the small, easily actuated release **118**. In this implementation the release **118** is hidden under kickstand **112**. This configuration can provide a technical solution of an easy release mechanism that is hidden to avoid accidental engagement. FIG. 6 shows an alternative configuration. The release **118** provides a technical solution that changes the state of the latch assembly **202** and allows the force or bias parallel to the rear side **106** to be conveyed to the rear cover **114** to move the rear cover parallel to the rear side. Further, the latch shuttle **316** can provide a technical solution that laterally conveys spring force across a distance. Thus, the latch assembly **202** can be positioned in the first area **108**, but convey a force on the rear cover **114** located in the second area **110**.

(27) When the user engages the release **118**, such as by pressing down with a pointed tool, such as a stylus, the rear cover **114** self-ejects from rear of device **100**. The rear cover **114** can be totally uniform on its exposed surface (e.g., no fasteners access ports, etc.). The underside of the rear cover can include hooks **332** that engage lips **402** on the chassis **102** to retain the rear cover **114**. When the hooks **332** disengage from the lips **402** due at least in part to lateral y direction movement, the plugs **328** are biased (e.g., in the z direction) toward the rear side **106** by ejection springs **324**. The plugs **328** provide a dual functionality technical solution of popping the rear cover **114** up from the chassis **102** so it can be readily removed by the user and the plugs **328** fill the cavity **330** at the rear side **106** to provide a self-closing feature. The self-closing feature prevents dirt/moisture ingress while the rear cover **114** is removed from device **100**.

(28) Reattachment or reinstallation of the rear cover **114** can be accomplished with two motions, press down and slide over. These motions are equivalent to the removal motion described with arrows **122** and **124** in FIG. 1C, except that the order and direction are reversed (e.g., slide and then

up to release versus down and then slide to re-install).

(29) Compared to other solutions such as fasteners or adhesive, the present concepts provide a faster and more convenient removal and reattachment experience of the rear cover by the user. There are no loose parts, such as screws that can be lost, and no sacrificial parts, such as adhesive tape that need to be replaced.

(30) The plugs **328** can also provide visual guidance for reinstallation of the rear cover **114**. The user can place the rear cover **114** on the chassis **102** so the hooks **332** are on the plugs **328** and the bumper **334** is positioned at the latch shuttle **316**. Once the rear cover **114** is aligned, the user can press down on the rear cover to overcome the bias of the ejection springs **324**, compress the ejection springs, and force the hooks **332** into the cavities **330**. The user can then slide the cover laterally into position so the hooks **332** are under the lips **402**. This lateral movement simultaneously causes the bumper **334** which is received in the latch shuttle **316** to move the latch shuttle **316** so that the shuttle hook **408** overrides and engages the latch hook **406** and the latch **314** resets and retains the latch shuttle **316** and via the latch shuttle, the rear cover **114**. This motion also compresses shuttle springs **308** and thus provides the energy for the automatic sliding of the rear cover when the release is depressed. At some subsequent point, the user can engage the release **118** to remove the rear cover **114** in a fraction of a second because the shuttle springs slide the rear cover **114** parallel to the rear side **106** and then the ejection springs **324** automatically pop the rear cover up and perpendicularly away from the rear side **106**. Thus, collectively, the shuttle springs **308** and the ejection springs **324** provide a technical solution that allows the user to re-install the rear cover **114** with a simple motion (e.g., down then sideways) rather than having to become proficient at an angled or curved installation path.

(31) FIG. **6** shows another example device **100A**. In this case, the chassis **102** defines the first area **108** of the rear side **106**. The second area **110** of the rear side **106** is covered by the rear cover **114**. The latch assembly **202** is positioned relative to the first area **108** and interacts with the rear cover **114** as explained above relative to FIGS. **1A-5B**. The release **118** is positioned on and accessed at the first area **108**. In this implementation, three ejection assemblies **204** secure the rear cover **114** to the rear side of the chassis **102**. Other numbers of ejection assemblies are contemplated. In these configurations, the latch assembly **202** can provide a technical solution for whether the device is in an operation condition or a service condition. If the user engages the release to service the device, the latch assembly **202** provides a spring force or bias to move the rear cover **114** parallel to the rear side **106**. This parallel movement can release the rear cover's hooks from the chassis and allow the ejection assemblies **204** to automatically bias the rear cover **114** away from the rear side **106** so the user can access the components in the chassis.

(32) The present concepts can be utilized with various types of user-serviceable devices, such as computing devices that can include, but are not limited to, notebook computers, tablet type computers, smart phones, wearable smart devices, gaming devices, entertainment consoles, and/or other developing or yet to be developed types of devices. As used herein, a computing device can be any type of device that has some amount of processing and/or storage capacity and/or other components. A mobile computing device can be any computing device that is intended to be readily transported by a user.

(33) Various examples are described above. Additional examples are described below. One example includes a device comprising a chassis defining a front side and a rear side, a kickstand that is rotatably secured to the rear side by a hinge that divides that rear side into a first area and a second area, the kickstand configured to fold against the first area for storage or be angled away from the first area to support the chassis, a rear cover positioned over the second area of the rear side to protect components positioned in the chassis, an ejection assembly positioned at the second area and configured to maintain the rear cover against the chassis, and a latch assembly positioned at the first area and comprising a release that is hidden by the kickstand when the kickstand is folded against the first area and that can be engaged through the first area when the kickstand is angled

away from the first area and engagement of the release is configured to cause the rear cover to move away from the kickstand substantially parallel to the rear side until released by the ejection assembly.

(34) Another example can include any of the above and/or below examples where the rear cover defines hooks that slideably engage lips defined by the chassis, and based at least on the rear cover movement away from the kickstand the hooks are configured to disengage from the lips.

(35) Another example can include any of the above and/or below examples where, based in part on the hooks disengagement from the lips, the ejection assembly is configured to bias the rear cover away from the rear side.

(36) Another example can include any of the above and/or below examples where the ejection assembly comprises ejection springs that bias plugs toward the rear cover.

(37) Another example can include any of the above and/or below examples where, based at least on the ejection assembly movement of the rear cover away from the chassis, the plugs are configured to fill cavities that were occupied by the hooks.

(38) Another example can include any of the above and/or below examples where the latch assembly defines a latch that is configured to rotate around a pivot rod and that is biased toward the rear side by a latch spring.

(39) Another example can include any of the above and/or below examples where the latch defines a latch hook.

(40) Another example can include any of the above and/or below examples where the latch assembly further comprises a latch shuttle that defines a shuttle hook and wherein the shuttle hook is configured to be captured by the latch hook when the latch is biased against the rear side.

(41) Another example can include any of the above and/or below examples where the latch shuttle is biased away from the latch by shuttle springs and wherein engagement of the release is configured to free the shuttle hook and allow the shuttle springs to move the shuttle away from the latch.

(42) Another example can include any of the above and/or below examples where wherein the latch shuttle engages the rear cover and is configured to move the rear cover away from the kickstand based in part on the shuttle springs movement of the shuttle away from the latch.

(43) Another example includes a device comprising a chassis defining a volume, a rear cover positioned against the chassis and providing access to the volume, an ejection assembly positioned on the chassis and configured to engage hooks on the rear cover to maintain the rear cover against the chassis over the volume, and a latch assembly positioned on the chassis and comprising a release that can be engaged through the chassis to cause a latch to pivot away from a latch shuttle to allow the latch shuttle and the rear cover to be moved by a shuttle spring until the hooks disengage from the chassis.

(44) Another example can include any of the above and/or below examples where the shuttle spring is compressed between the latch and the latch shuttle or wherein the shuttle spring is compressed between the chassis and the latch shuttle.

(45) Another example can include any of the above and/or below examples where the latch shuttle engages a bumper on an underside of the rear cover.

(46) Another example can include any of the above and/or below examples where the ejection assembly comprises ejection springs that bias the rear cover substantially perpendicular to the bias of the shuttle springs.

(47) Another example can include any of the above and/or below examples where ejection springs contact plugs and bias the plugs against the hooks and wherein based in part on the rear cover movement away from the ejection springs the plugs fill a cavity that was occupied by the hooks.

(48) Another example can include any of the above and/or below examples where no fasteners extend through or are exposed on the rear cover when the rear cover is installed on the chassis over the cavity.

(49) Another example includes a device comprising a chassis defining a volume accessed through a rear side, a rear cover positioned against the chassis over the rear side, an ejection assembly positioned on the chassis and configured to allow movement between the rear cover and the chassis substantially parallel to the rear side to lock the rear cover against the chassis and to allow movement in an opposite direction to unlock the rear cover and further configured to impart a biasing force on the rear cover substantially perpendicular to the rear side, and a latch assembly positioned on the chassis and comprising a release that can be engaged through the chassis to cause the movement of the rear cover in the opposite direction to unlock the rear cover and allow the biasing force to move the rear cover substantially perpendicularly to the rear side away from the chassis.

(50) Another example can include any of the above and/or below examples where the ejection assembly is positioned in a cavity in the chassis and wherein the ejection assembly comprises a plug that is biased against the rear cover.

(51) Another example can include any of the above and/or below examples where the plug fills the cavity at the rear side based in part on the rear cover removal.

(52) Another example can include any of the above and/or below examples where the ejection assembly comprises an ejection assembly at each corner of the rear cover.

(53) Conclusion

(54) Although techniques, methods, devices, systems, etc., pertaining to user-serviceable devices are described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed methods, devices, systems, etc.

Claims

1. A device, comprising: a chassis defining a front side and a rear side; a kickstand that is rotatably secured to the rear side by a hinge that divides that rear side into a first area and a second area, the kickstand configured to fold against the first area for storage or be angled away from the first area to support the chassis; a rear cover positioned over the second area of the rear side to protect components positioned in the chassis; an ejection assembly positioned at the second area and configured to maintain the rear cover against the chassis; and, a latch assembly positioned at the first area and comprising a release that is hidden by the kickstand when the kickstand is folded against the first area and that can be engaged through the first area when the kickstand is angled away from the first area and engagement of the release is configured to cause the rear cover to move away from the kickstand substantially parallel to the rear side until released by the ejection assembly.
2. The device of claim 1, wherein the rear cover defines hooks that slideably engage lips defined by the chassis, and based at least on the rear cover movement away from the kickstand the hooks are configured to disengage from the lips.
3. The device of claim 2, wherein based in part on the hooks disengagement from the lips, the ejection assembly is configured to bias the rear cover away from the rear side.
4. The device of claim 3, wherein the ejection assembly comprises ejection springs that bias plugs toward the rear cover.
5. The device of claim 4, wherein based at least on the ejection assembly movement of the rear cover away from the chassis the plugs are configured to fill cavities that were occupied by the hooks.
6. The device of claim 1, wherein the latch assembly defines a latch that is configured to rotate around a pivot rod and that is biased toward the rear side by a latch spring.
7. The device of claim 6, wherein the latch defines a latch hook.

8. The device of claim 7, wherein the latch assembly further comprises a latch shuttle that defines a shuttle hook and wherein the shuttle hook is configured to be captured by the latch hook when the latch is biased against the rear side.
 9. The device of claim 8, wherein the latch shuttle is biased away from the latch by shuttle springs and wherein engagement of the release is configured to free the shuttle hook and allow the shuttle springs to move the shuttle away from the latch.
 10. The device of claim 9, wherein the latch shuttle engages the rear cover and is configured to move the rear cover away from the kickstand based in part on the shuttle springs movement of the shuttle away from the latch.
 11. A device, comprising: a chassis defining a volume; a rear cover positioned against the chassis and providing access to the volume; an ejection assembly positioned on the chassis and configured to engage hooks on the rear cover to maintain the rear cover against the chassis over the volume; and, a latch assembly positioned on the chassis and comprising a release that can be engaged through the chassis to cause a latch to pivot away from a latch shuttle to allow the latch shuttle and the rear cover to be moved by a shuttle spring until the hooks disengage from the chassis, the latch shuttle being configured to travel in a linear direction towards and against the chassis by elongation of the shuttle spring from a compressed configuration.
 12. The device of claim 11, wherein the shuttle spring is compressed between the latch and the latch shuttle or wherein the shuttle spring is compressed between the chassis and the latch shuttle.
 13. The device of claim 12, wherein the latch shuttle engages a bumper on an underside of the rear cover.
 14. The device of claim 13, wherein the ejection assembly comprises ejection springs that bias the rear cover substantially perpendicular to the bias of the shuttle springs.
 15. The device of claim 14, wherein the ejection springs contact plugs and bias the plugs against the hooks and wherein based in part on the rear cover movement away from the ejection springs the plugs fill a cavity that was occupied by the hooks.
 16. The device of claim 15, wherein no fasteners extend through or are exposed on the rear cover when the rear cover is installed on the chassis over the cavity.
 17. A device, comprising: a chassis defining a volume accessed through a rear side; a rear cover positioned against the chassis over the rear side; an ejection assembly positioned on the chassis and configured to allow movement between the rear cover and the chassis substantially parallel to the rear side to lock the rear cover against the chassis and to allow movement in an opposite direction to unlock the rear cover and further configured to impart a biasing force on the rear cover substantially perpendicular to the rear side; and, a latch assembly positioned on the chassis and comprising a release that can be engaged through the chassis to cause the movement of the rear cover in the opposite direction to unlock the rear cover and allow the biasing force to move the rear cover substantially perpendicularly to the rear side away from the chassis, the latch assembly includes a shuttle spring configured to provide a first biasing force for the parallel movement and an ejection spring configured to provide a second biasing force for the perpendicular movement, the shuttle spring being compressed in a direction parallel to the rear side when the rear cover is in a locked position.
 18. The device of claim 17, wherein the ejection assembly is positioned in a cavity in the chassis and wherein the ejection assembly comprises a plug that is biased against the rear cover.
 19. The device of claim 18, wherein the plug fills the cavity at the rear side based in part on the rear cover removal.
 20. The device of claim 19, wherein the ejection assembly comprises an ejection assembly at each corner of the rear cover.
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